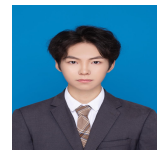


GE Hanze

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Target: Equity & Industry Research (Energy Technology • Auto Value Chain • AI • New Materials • Smart Manufacturing)



Profile: Chemical Engineering & Technology (National First-Class Programme) with a technology-plus-finance profile: tracks five sectors on a policy–technology–supply & demand–fundamentals framework and tests disclosed capacity, cost and technology-barrier assumptions against plant-floor and lab evidence; builds material/energy balances and Excel cost models.

EDUCATION

Northeast Petroleum University | Chemical Engineering & Technology (National First-Class Programme) | BSc, Full-time Sep 2022 – Jun 2026

Core courses: Chemical Engineering Principles | Thermodynamics | Reaction Engineering | Polymer Materials | Process Design | Aspen Plus | EHS

• **Honours & awards:** East Money Financial Challenge – Provincial First Prize (top 5%) | Financial Innovation Practice Competition – Provincial Second Prize | Futures Simulation Competition – University First Prize | National English Competition – Second Prize | Computer Rank Level 2 | CAD Level 2

RESEARCH & INVESTMENT ANALYSIS

Independent Sector Deep-Dives + Competition Practice | Core Contributor 2021 – Present

- **Two sector deep-dive reports, sole author:** CF/PEEK engineering plastics: sized TAM across humanoid robotics, 800V EV motors and low-altitude aviation and weighed three modification routes on performance, process and cost; CVD Si/C anode: decomposed porous-carbon scaffold and fluidised-bed barriers, modelling per-tonne cost (c. RMB 100k/t, 33%–44% margin) with silane-price sensitivity.
- **Five-sector value-chain tracking:** Track energy technology (battery materials/storage), auto (compounds/coatings), AI (compute), advanced materials (carbon/fluorine) and smart manufacturing (extrusion/compounding) on a policy–technology–supply & demand–fundamentals framework, publishing value-chain maps and supply/demand calls feeding screening.
- **Primary-source edge and quantitative validation:** Provincial and university first prizes in the East Money Financial Challenge (top 5%, telecom and AI-compute research) and simulated portfolios above 20%; cross-checks disclosed capacity, utilisation and process-cost assumptions against plant-floor evidence.

WORK & INTERNSHIP EXPERIENCE

Leistritz Machinery (Taicang) Co., Ltd. | Process Engineer (Technical Application & Field Support) Jul 2026 – Present

- **Value-chain mapping:** Serve six leading accounts — ExxonMobil, CATL, LyondellBasell, Daikin, Kingfa, Changchun Chemical — across polyolefins, flame retardants, masterbatch, carbon electrodes and fluorine-free systems.
- **Process-window modelling and root-cause analysis:** Built a material / screw-configuration / parameter sensitivity matrix on twin-screw extrusion (ZSE27CC, 30–100 kg/batch) and led root-cause analysis on five field-fault classes — surging, torque, melt-temperature and pressure drift — moving key quality indicators from rejected to released.
- **Technology-route assessment:** Hands-on with PFAS-free, flame-retardant and recycled/bio-based systems; judge substitution feasibility from the processing-stability, performance and cost gap versus fluorinated routes.

Synthomer Shanghai Co., Ltd. | Coatings R&D & Technical Application Intern Jul 2025 – Oct 2025

- **Reverse-engineering the requirement:** Co-developed a waterborne anti-chip automotive underbody coating with Henkel China, converting OEM adhesion, impact and application-window requirements into formulation targets and a validation matrix.
- **Quantified optimisation:** Led slurry dispersion optimisation: viscosity down about 10%, impact resistance up about 15% at lab scale, quantifying the performance–cost trade-off boundary.

PetroChina (Daqing Petrochemical No.2 Refinery / Daqing Oilfield Light-Hydrocarbon Fractionation) | Field Process Intern Jun 2025 – Jul 2025 | Jul 2024 – Sep 2024

- **High-frequency data tracking and supply-side insight:** 200+ inspection rounds and 100+ operating-data sets, ruling out 10+ equipment abnormalities; understand how load–energy–yield constraints drive utilisation, maintenance cycles and product mix.
- **Value-chain flows and research output:** Tracked the ethylene/propylene/butene fractionation train, optimised three operating parameters and mapped C3/C4 flows and downstream pricing logic.

KEY RESEARCH & ENGINEERING PROJECTS

Modified Biomass-Derived Carbon Electrodes for Supercapacitors | Core Member Sep 2025 – Jun 2026

- **Maturity validation:** Prepared and characterised biomass-derived electrodes by green activation (SEM/XRD/BET, 2143.6 m²/g), 92% capacitance retention over 5,000 cycles — maturity evidence for biomass hard-carbon in sodium-ion and storage.

Dimethyl Sulfoxide (DMSO) Clean-Production Plant Process Design | Project Lead Feb 2025 – Jul 2025

- **Flow simulation and economics:** Aspen Plus full-flow simulation, material/energy balances, PFD/P&ID and GB50160 layout; parameter optimisation raised raw-material use 18% and cut by-products 23%, with unit-cost and capacity assessment.

SKILLS & QUALIFICATIONS

- **Investment research:** Value-chain mapping | Competitive structure (CRn) | Route evolution and substitution | Supply/demand balance and utilisation tracking | Unit economics and cost curves
- **Modelling & analysis:** Material/energy balances | Cost structure and sensitivity analysis | Scenario and stress testing | Excel modelling | Aspen Plus process economics
- **Tools & languages:** Excel (modelling/pivots) | PowerPoint (research layout) | Aspen Plus | Public market data platforms (East Money) | CET-6 | TOEFL (CEFR B2)
- **Regulatory literacy (in progress):** Exchange trading rules | Listed-company refinancing | Disclosure and inside-information management